Mukul Kumar

Software Engineer — C++ & Python — Scalable Data Architectures

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Education

Indian Institute of Technology Hyderabad 2022 – 2026
B.Tech – Chemical Engineering GPA: 7.14

R.K.C Vidya Vihar Bohatwas Ahir, Rewari 2021 – 2022
Class XII

R.K.C Vidya Vihar Bohatwas Ahir, Rewari 2019 – 2020
Class X

Skills

- **Programming Languages & Frameworks:** Python, C++, MATLAB, FastAPI
- **CS Fundamentals:** Data Structures, Algorithms, Operating Systems, DBMS, OOPs
- **Distributed & Scalable Systems:** Docker, Linux Networking, gRPC, Protobuf, Semantic Search
- **Software Development Practices:** Teamwork, Communication, System Testing, Problem Solving
- **Data & AI Concepts:** Embeddings, Time Series Forecasting, TensorFlow, scikit-learn

Work Experience

People Tech Group Inc 06/2025 – 07/2025
SDE Intern

- Developed a high-performance C++ middleware bridge to enable real-time data transfer from legacy SOME/IP-based ECUs to the KUKSA Databroker, supporting scalable data architecture for software-defined vehicles.
- Applied knowledge of C++, gRPC, Protobuf, and Linux networking to design and test robust backend systems, optimizing for reliability and performance.
- Built and maintained the VSOMEIP adapter, implementing advanced CAN signal parsing and translating binary payloads into Protobuf messages to ensure type safety and cross-platform compatibility.
- Leveraged database management principles to enhance data integrity and support efficient data flow within distributed automotive systems.
- Collaborated closely with product management and design teams to align middleware features with user requirements and improve overall system usability.

Projects

DocuMind AI – Document Intelligence Platform
AI-powered platform enabling PDF upload, semantic search, and audio insights via Podcast Mode.

- Ranked in the Top 100 out of 60,000+ teams at Adobe Hackathon.
- Designed and implemented FastAPI backend with APIs for document ingestion and semantic queries.
- Engineered an end-to-end Azure TTS pipeline for automated audio generation in Podcast Mode.
- Integrated Google Gemini, OpenAI embeddings, and Docker for scalable, modular deployment.

Smart Expense Splitter System
High-performance backend platform for managing group expenses with optimized settlements.

- Developed a scalable backend using **Java 17 and Spring Boot**, modeling a highly normalized MySQL database via **Spring Data JPA** to handle complex Many-to-Many relationships.
- Engineered a **Greedy Settlement Algorithm** to compute net balances and optimally match debtors with creditors, minimizing total transaction count.
- Optimized data pipelines by resolving **bidirectional ORM dependency issues**, ensuring stable JSON serialization and faster REST API responses.

Graph-Based Recommendation System
High-performance recommendation engine modeling user-item interactions as weighted graphs with real-time updates.

- Developed in C++17 with multi-metric similarity (Cosine, Jaccard, Pearson, Euclidean) and priority-queue traversal for top-K recommendations.
- Optimized system for sub-millisecond latency to support real-time recommendations.

Certificates & Achievements

- NTSE Scholar: Qualified Stage-I & Stage-II (2020–2021)
- Qualified UPSC – NDA
- NCC C-Certificate with 'A' Grade
- Part of the INTER IIT Volleyball Team